

Business Data for Research Guidance Document

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Version Control

Version	Date	Changes Made	Dataset Version	Owner	Contact
1.0	March 2023	Initial release of the BDR User Guide	2014–2021	NISRA ELMS	rsu@nisra.gov.uk
2.0	October 2025	Added 2022 data, new exports variables, ASHE dataset included	2014–2022	NISRA ELMS	rsu@nisra.gov.uk

1 Introduction

This document provides guidance to researchers on the Business Data for Research (BDR) database. The BDR is a longitudinal de-identified themed dataset including NISRA business data from 2014-2022 in Northern Ireland.

The database has been created through the ADR NI using the Digital Economy Act 2017 as the legal gateway. The directive of the ADR NI has been to provide a secure and accredited research environment to enable agreed, ethically approved and cross-cutting research to be conducted using administrative data sets that are routinely collected and analysed by NICS departments and other bodies. The ultimate aims of the joined-up approach are to inform the development and monitoring of public policy and to help ensure that decision making is evidence-based and the research provides a benefit to society. The ADR NI makes it possible for trained researchers to use **pseudonymised** administrative data for social and economic research, while making sure the data remain safe and the **privacy of individuals and businesses are protected**.

Researchers should note that NISRA has strict disclosure rules in place when analysing our data as we are bound by the Code of Practice for Statistics, and the legislation under which we collect the data i.e. The Statistics of Trade and Employment (NI) Order 1988, to protect the identity of businesses who provide their information to us. Therefore, we must ensure that the identity of businesses that provide information to us is protected at all times. This means that it is not possible to make certain variables available to researchers such as business address or postcode. It also dictates the disclosure rules that determine what can and cannot be published from analysis of our data.

The Code of Practice for Statistics also places a duty on NISRA to minimise business burden when responding to statutory surveys. This requirement has implications for matters such as the sample size of a survey and the length of a survey.

Included in this document are background information on the data sources that constitute the BDR, a description of the BDR and how researchers can access the data.

Also included is a description of variables and caveats that are essential for the understanding of the BDR structure. When applying to use the BDR data this document should be used in conjunction with this BDR metadata and data dictionary.

The initial pilot of the development of the BDR database was funded by the ESRC and completed in 2022/23. Seven beta projects were undertaken to test the database and associated documentation available for researchers. Feedback from the researchers has been incorporated into the development of the database, this guidance documentation and the accompanying documentation available to researchers.

The database will be subject to various structure changes as time progresses which can mostly be attributed to the addition of various data over time. These changes will be reported in future versions of this guidance document. This is the second edition of this document. Changes since the first edition are the inclusion of 2022 data for the existing BDR datasets, the inclusion of new exports variables and the addition of the Annual Survey of Hours and Earnings (ASHE) dataset 2014-2022.

The owner of this document is NISRA Economic and Labour Market Statistics Branch (ELMS) and any changes to the document should be suggested to NISRA-ELMS who will make amendments if required. Please contact NISRA Research Support Unit (RSU) in the first instance (rsu@nisra.gov.uk).

1.1 Data Collection, Validation, Imputation Processes & Revisions

1.1.1 Data collection

All of the NISRA business surveys featured in the BDR are administered by ELMS within NISRA.

To administer and collect the data ELMS utilise an advanced data collection system which facilitates data collection via online and paper formats.

1.1.1.1 Sampling Frame - Inter Departmental Business Register

The Inter Departmental Business Register (IDBR) is the sampling frame used for all NISRA business surveys¹. The register consists of companies, partnerships, sole proprietors, public authorities, central government departments, local authorities and non-profit making bodies in the UK.

The IDBR covers most of the economy including the Agriculture, Production, Construction and Service sectors in Northern Ireland. It does not, however, include very small businesses which are not VAT registered (the threshold for VAT in 2022/23 was £85,000) unless they operate a PAYE scheme. This means that many self-employed workers will not be included on the IDBR.

In 2023, the IDBR estimates that number of VAT and/or PAYE registered businesses operating in Northern Ireland is 79,235.

However, the [Department for Business and Trade \(DBT\)](#) estimates (Table 23) that in 2023 there were approximately 47,295 unregistered businesses in Northern Ireland, which is in addition to the figures reported by the IDBR.

(Please note the definitional difference: DBT and ONS use count of enterprises from the IDBR whereas NISRA use count of businesses operating in NI.)

The business activity, or industry, is recorded using Standard Industrial Classification (UK SIC 2007) codes.

Statistical units are classified using a hierarchy which is described below.

- **Enterprise Group:** is a group of legal units under common ownership. Enterprise is the smallest combination of legal units (generally based on VAT and/or PAYE records) which has a certain degree of autonomy within an Enterprise Group. The geographical location of an

¹ Note that while the IDBR is used to obtain the contact details of employers of individuals sampled within the ASHE, the ASHE sample is drawn from HM Revenue and Customs' (HMRC's) Pay as you Earn (PAYE) system. Please see section 1.1.1.2.

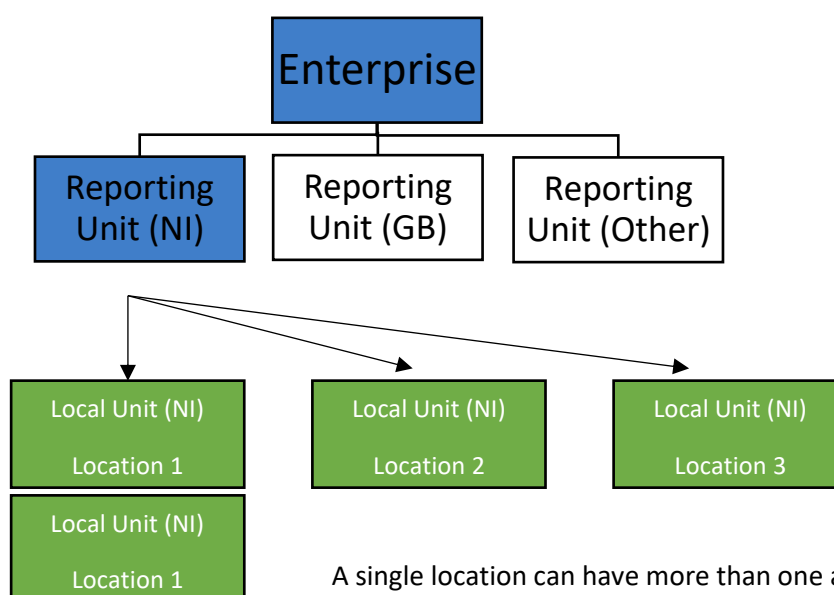
Enterprise is based on its registered (or main) UK address. Note that no datasets in the BDR have any variables relating to Enterprises.

- **Reporting Unit:** includes all or part of an Enterprise. There will be at least one Northern Ireland Reporting Unit for any business operating in Northern Ireland.
- **Local Unit:** is an individual site (for example a factory or shop). Usually all sites in Northern Ireland are included in a single Reporting Unit. A Northern Ireland Reporting Unit will not include any sites outside Northern Ireland.

The ONS defines a business as an enterprise and assigns each business to a geographical location based on its registered or main address, within the UK. This is the standard EU definition but has the disadvantage that the Northern Ireland figures will not include any businesses operating in Northern Ireland, which have their main or registered address elsewhere in the UK. An alternative approach is to define each Northern Ireland Reporting Unit as a business. This ensures that all businesses operating in Northern Ireland are included but has the drawback that any Enterprise with more than one Reporting Unit in Northern Ireland will be counted as multiple businesses.

To address these issues an Enterprise operating within Northern Ireland definition has been introduced. Under this definition each Enterprise operating in Northern Ireland is included in the Northern Ireland figures but only once and the industrial classification, employment/employees and turnover information only relate to the part of the business located in Northern Ireland. Where the Enterprise has more than one Northern Ireland Reporting Unit the data for the individual Reporting Units are aggregated to form a composite unit. The industrial classification for the composite unit is determined based on the industrial classifications of the local units contained within the unit, using the same method as is used for a standard Reporting Unit. Where a business operates across multiple SICs at different local units, the dominant SIC across LUs as determined by employment is applied to the RU. This also applies when combining RUs to create a composite unit. The location of the composite unit is determined based on the address of the site with overall responsibility for Northern Ireland activity or the primary operating site within Northern Ireland.

The relationship between enterprises, reporting unit and local unit is depicted below.



A single location can have more than one activity.

1.1.1.2 Sampling Frame - HM Revenue and Customs Pay As You Earn system

The ASHE survey uses a random sample of 1% of all employee jobs from HM Revenue and Customs's (HMRC's) Pay As You Earn (PAYE) system, taken in January of the reference year. The sample is drawn in such a way that many of the same individuals are included from year to year, thereby allowing longitudinal analysis of the data.

The sample is matched against the IDBR in order to obtain contact and address details for the employers. Information on the hours paid and earnings of employees is obtained from employers and treated confidentially.

The survey has a reference date in April and asks about individuals who were employees at that time. The reference date changes each year depending on when Easter falls. A second extract is taken from the PAYE system in April in order to identify people who have either joined the labour market or changed jobs since the January sample was taken.

Since ASHE is a survey of employee jobs, it does not cover the self-employed or any jobs within the armed forces. Given the survey reference date in April, the survey does not fully cover certain types of seasonal work, for example, employees taken on for only summer or winter work.

Statistical units included within the ASHE are described below.

- **Jobs:** individuals within the ASHE can have multiple jobs. The number of jobs can be ascertained by a count of rows per year.
- **Individuals:** The number of individuals can be ascertained by a count of the variable Anonymised_piden per year.
- **Reporting Unit:** includes all or part of an Enterprise. There will be at least one Northern Ireland Reporting Unit for any business operating in Northern Ireland. The number of reporting units can be ascertained by a count of the variable Anonymised_RURef per year.

1.1.2 Data validation

When data is submitted by businesses there are a range of comprehensive validation rules in place to ensure that data provided is sound and within expected ranges. Where a data point falls outside of an expected range or fails a validation rule, a task is created whereby a member of staff will check the data and where appropriate contact the business to confirm the value and receive an explanation from the business to support the reason for the larger than expected change in the value.

These processes ensure that the data we analyse for our results production are robust, sound and fit for their originally intended purpose.

See [Annex E](#) for data quality reports.

1.1.3 Imputation

For NIABI and BRES, NISRA utilises established and robust imputation processes to estimate values for variables where possible, for businesses who have not made a return in the current year. This includes

in the census component of survey samples. The processes are aligned with best practice and are equivalent to the processes adopted by ONS for similar business surveys.

The imputation process draws upon a range of data including previous returned values by a business, values for respective variables submitted by other similar businesses in the same detailed industry classification and size band, the Inter Departmental Business Register (IDBR) (further information on the IDBR is detailed below) and other internal data sources held by ELMS. As such, the imputation processes are considered robust.

For ASHE, Donor imputation is carried out for missing data items on partial returns; no non-responder imputation is carried out.

There are no imputation markers available on the datasets.

1.1.4 Revisions

In most business surveys, the most recent data published is provisional and are subject to revision in the next reporting period.

In an attempt to limit the costs to those who complete statistical returns, the majority of economic and labour market statistics are estimates of what they are measuring, rather than being complete counts. Reliance on surveys to collect information with which to produce statistics means that events and socio-economic trends cannot be measured in their entirety, but are instead based on incomplete information which is adjusted to take account of biases which arise in survey taking, and other available information, such as the emergence of new seasonal patterns.

Over time, as more information becomes available, estimates can be revised to improve quality and accuracy, which will provide a better picture of what is being measured. Users require good quality data that are timely and can be accessed as close to the event which the data measures as possible. A policy of accepting revisions enables statisticians to provide users with data close to the event, but also improve the accuracy of that data as other information becomes available. Revisions are therefore a standard practice when producing official statistics.

All data within the BDR are finalised. See [Annex E](#) for revisions policies.

1.2 Quality Assurance

Quality assurance practices were applied by ELMS to the BDR datasets to ensure the data was accurate, reliable, and representative. RSU completed a statistical disclosure control assessment of the data.

1.2.1 Consistency within the data

ELMS checked the data for anomalies, unexpected values, and issues with aggregations. Where issues were identified these were relayed to the survey teams, and amended based on the advice received. For example, some of the SIC variables didn't have a leading zero, which had to be added. Some variables were blank or had negative values. These were investigated and most were deemed to be correct and not changed.

Some internal inconsistencies were highlighted, where sums would not add to totals. While these were accepted as differences were very small and shouldn't have an impact, ELMS are continually monitoring these and actively seeking ways to reduce the instances of inconsistencies within the data.

Details of the minor internal inconsistencies in the Northern Ireland Annual Business Inquiry (NIABI) data were:

WQ059: Formula of [WQ050]+[WQ055]+[WQ058] does not always sum to WQ059

WQ400: Formula of [WQ412]+[WQ413]+[WQ415]+[WQ431] does not always sum to WQ400

WQ499: Formula of [WQ710]+[WQ720]+[WQ730] does not always sum to WQ499

WQ550: Formula of [WQ399]+[WQ346]-([WQ321]+[WQ312]+[WQ313]+[WQ314]) does not always sum to WQ550

WQ600: Formula of

[WQ610]+[WQ611]+[WQ622]+[WQ612]+[WQ614]+[WQ615]+[WQ626]+[WQ627]+[WQ629]+[WQ643]+[WQ644] does not always sum to WQ600

WQ690: Formula of [WQ600]-[WQ699] does not always sum to WQ690

WQ699: Formula of

[WQ630]+[WQ631]+[WQ636]+[WQ637]+[WQ638]+[WQ639]+[WQ641]+[WQ642]+[WQ645] does not always sum to WQ699

WQ720: Formula of [WQ732]+[WQ729] does not always sum to WQ720

WQ729: Formula of [WQ721]+[WQ722]+[WQ723]+[WQ724]+[WQ725]+[WQ726]+[WQ727] does not always sum to WQ729

WQ730: Formula of [WQ780]+[WQ770]+[WQ771]+[WQ772]+[WQ1830]+[WQ734]+[WQ775] does not always sum to WQ730

WQ2010: Formula of [WQ550]+[WQ317]+[WQ599]+[WQ502]-[WQ500]-[WQ501]+[WQ602]-[WQ499] does not always sum to WQ2010

WQ2020: Formula of [WQ2010]+[WQ414]-[WQ400]+[WQ416] does not always sum to WQ2020

WQ2030: Formula of [WQ2020]+[WQ412]+[WQ431] does not always sum to WQ2030

One of the beta testing research teams also highlighted that the Total exports of goods & services outside UK does not always sum to export goods outside UK + export services outside UK.

Investigation showed they sum the individual parts instead, i.e. exports ROI goods + exports ROI services + exports REU goods + exports REU services + exports ROW goods + exports ROW services = Total exports goods & services outside UK.

The same methodology applies to Imports, i.e. imports ROI goods + imports ROI services + imports REU goods + imports REU services + imports ROW goods + imports ROW services = Total imports goods & services outside UK.

For Exports, in variable terms, Total exports goods & services outside UK = [WQ114]+[WQ111]+[WQ113]+[WQ116]+[WQ115]+[WQ117]

For Imports, in variable terms, Total imports goods & services outside UK = [WQ124]+[WQ121]+[WQ123]+[WQ126]+[WQ125]+[WQ127]

It should be noted that some minor inconsistencies occur within the ASHE dataset in relation to sex. Changes to the recording of sex across or within years are likely due to collection errors and researchers are advised to treat these as such.

1.2.2 Consistency with published reports

ELMS sought to replicate the headline statistics for each of the subject datasets. This was to ensure that the datasets provided are an accurate reflection of the data used internally by ELMS to produce the respective publications. Where discrepancies were identified these were investigated by the ELMS teams and remedial action taken to ensure the final BDR datasets are consistent with published reports.

1.2.3 Statistical Disclosure Control (SDC)

RSU worked with ELMS to ensure that the data provided was safe and that it could not be used to disclose small numbers. For instance, feedback from researchers in the pilot phase included the desire to incorporate geographical markers such as post codes or Super Output Area markers.

RSU and ELMS undertook disclosure testing to determine whether or not this would be viable. Given the small size of the NI economy and business population, it was determined that providing SOA markers, would likely cause disclosure issues.

ELMS have strict disclosure rules in place as we are bound by the Code of Practice for Statistics, and the legislation, to protect the identity of businesses who provide their information to us. These same rules are applied to all of our business statistics and associated outputs. As such these geographical variables cannot be made available.

However, we have agreed to include Local Government District markers on the datasets as this is considered to be the safest geographical identifier available.

Researchers should also note that, following an SDC assessment of the ASHE dataset, the Age variable has been bottom-coded at age 17 and top-coded at age 67.

2 BDR datasets

The BDR is comprised of four large scale annual datasets which are highlighted below and with further information available in subsequent sections.

1. Northern Ireland Annual Business Inquiry (NIABI);
2. Northern Ireland Economic Trade Statistics (NIETS);
3. Northern Ireland Business Register and Employment Surveys (BRES); and
4. Annual Survey of Hours and Earnings (AHSE).

2.1 Northern Ireland Annual Business Inquiry (NIABI)

The Northern Ireland Annual Business Inquiry (NIABI) is conducted on a statutory basis under the Statistics of Trade and Employment (Northern Ireland) Order 1988 (Article 8). It collects both financial and employment information from businesses and other establishments and covers about two thirds of the economy. This includes the production, construction, distribution and service industries in Northern Ireland but excludes the financial sector and public sector activity for the most part.

The NIABI is a National Statistic² dataset and surveys all businesses with 50 or more employees; all multi-site businesses with 20-49 employees; all manufacturing businesses with more than 5 employees; and a representative sample of other smaller businesses. In all, the ABI samples about 20% of the eligible business population each year.

The NIABI provides a number of high-level indicators of economic activity such as the total value of sales and work completed by businesses (Turnover), the value of the purchase of goods, materials and services and total employment costs. The contribution of different industries to the overall value of economic activity can be assessed and because estimates of employment are collected at the same time it is also possible to get a measure of value added and costs per head to allow better comparison between different sized industrial sectors. In addition, data used to produce a number of trade related measures within the Northern Ireland Economic Trade Statistics (NIETS) are derived from the NIABI survey (see below for further information).

The NIABI is designed to provide the best estimates for Northern Ireland as well as providing information that is used at a later stage to inform UK National and Regional Accounts estimates.

The business unit to which NIABI questionnaires are sent is called the reporting unit. The response from the reporting unit can cover the enterprise as a whole, or parts of the enterprise identified by lists of local units. Other than for a minority of larger businesses which have a more complex structure, the reporting unit is the same as the enterprise. An enterprise may consist of one or more sub-units (called local units), for example, the head office for a group of shops. An enterprise may therefore have local units at different locations, and may carry out more than one type of economic activity. ELMS publish findings from the NIABI at both reporting (enterprise) and local unit levels. However, data is only available at reporting level.

² National Statistics status means that our statistics meet the highest standards of trustworthiness, quality, and public value, and it is our responsibility to maintain compliance with these standards. National Statistics accreditation is granted by the UK statistics Authority following comprehensive and independent assessment of the statistics.

Please note that the data within the NIABI and NIETS datasets are based on Reporting Unit (i.e. head office) information which means that all activity is coded based on the classification and location of the reporting unit. However, in reality, a business may have multiple sites or indeed a dedicated transport/logistics site from which goods are transported.

Researchers should, therefore, be aware that any analysis at Local Government District (LGD) level may be impacted by a “Head Office” effect. Researchers should also note that in some instances it is not possible to assign a business to a postcode. This is usually because the head office is outside NI. In such cases LGDs are labelled as ‘Unknown’ and hence summing the values for each LGD may not equal published NI totals. NIABI data is available on the BDR from the 2014 reference year up to 2022.

Salient points to note around the dataset include:

- **Sampling Frame**

The Inter Departmental Business Register (IDBR) is the sampling frame used for all NISRA business surveys, including the NIABI. The register consists of companies, partnerships, sole proprietors, public authorities, central government departments, local authorities and non-profit making bodies in the UK.

To maximise survey precision, the Neyman allocation approach to sampling is utilised. The survey universe is stratified by 2 digit SIC code and employee size band, and all businesses with 50+ employees, or 20+ employees and more than one local unit, are fully enumerated. Businesses falling below the threshold of complete enumeration are selected on a random stratified basis.

NIABI and NIETS sample size and returns from 2014 to 2022 are as follows:

Year	2014	2015	2016	2017	2018	2019	2020	2021	2022
NIABI (including NIETS)									
Record numbers (sample size/ returns)	10,000 approx./ 7,445	9,703/ 6,999	8,913/ 6,044	9,310/ 5,713	9,233/ 6,409	9,687/ 5,770	9,968/ 6,664	10,893/ 6,034	10,251/ 6,614

- **Stratification of the ABI sample**

The sample design for NIABI selects businesses based on selected employees within industry section at a **Reporting Unit level**.

A desired level of **estimate precision is set by industry section** and then the survey universe is stratified by the SIC2007 classification (see below) and then a Neyman Allocation apportionment at the **2 digit sic/size-band** level of the section totals is applied. This is a statistical process to

determine the optimal number of businesses that should be sampled in each strata to produce accurate and precise population level estimates.

Sample size bands are zero, 1-5, 6-9, 10-19 and 20-49 (part sample, part census - see below).

- **Sample size**

The NIABI sample ranges between 9,000-11,000 businesses per year out of approximately 55k survey-eligible businesses. This is a huge sample for a complex business survey of this nature. By way of example, NIABI covers approximately 18% of the eligible business population whereas the equivalent ONS ABS covers approximately 3% of the UK eligible business population.

- **Classifications**

The UK Standard Industrial Classification of Economic Activities 2007 (SIC 2007) is the classification framework used to define the industry in which a business operates. This is used by all business surveys within the UK.

- **Inclusions and exclusions**

NIABI covers the production, construction, distribution and service industries in Northern Ireland but excludes public sector activity for the most part. Coverage also includes agriculture (support activities), hunting, forestry and fishing.

Along with those exclusions stated above in section 2.1, the other main areas that are excluded are public administration and defence (section O) and Financial Industries (SIC 64-66) while agriculture, forestry and fishing (section A) excludes farming (groups 01.1 - Growing of non-perennial crops, 01.2 - Growing of perennial crops, 01.3 - Plant propagation, 01.4 - Animal production and 01.5 - Mixed farming). Local authority and central government bodies in education (section P) and human health and social work activities (section Q) have also been excluded from this publication, as has 86.2 (medical and dental practice activities) within section Q.

- **Blanks And Missing data**

The ABI weighting process is extremely complex and requires specialist software to undertake this. This software will not allow non-zero values or blank/missing variables to exist or else the algorithms will not work. Therefore, all cells in the database are numerated **which means that it is not possible to leave a cell blank** if a business did not provide data for a particular variable. Therefore, care is required when using the data for unit/firm level analysis as some zeros could actually be blanks.

- **Weights**

The use of weights within survey data is important to help ensure that the data is representative of the overall population. Users should note that the NIABI dataset includes weight variables, which have been calculated specifically for the sample drawn for each survey cycle, and used in the production of the NIABI published results. When replicating the published NIABI figures it is important that the correct weight is used, i.e. the “calweight”.

Valid sample returns are weighted to ensure they reflect the business population that are reported on.

The final weight, calweight (w), that is applied to each return is produced based on a design 'a' weight, and a calibration 'g' weight.

Let y_i = the value of the characteristic (e.g.: turnover) for the i th sampled business
 w_i = the weight for the i th sampled business
 $\bar{Y}$ = the desired population total
 $\hat{Y}$ = the estimate of $\bar{Y}$

Then:

$$\hat{Y} = \sum_i w_i y_i$$

The weight w_i is commonly split into three parts:

$$\hat{Y} = \sum_i a_i g_i o_i y_i$$

where a_i = the a-weight for business i

g_i = the g-weight for business i

o_i = the outlier weight for business i (For NIABI, this can be ignored and considered to have value = 1)

The design or 'a' weight is a simple expansion estimator (i.e: $\frac{N}{n}$) for similar groups in the population. So, for example, each SIC or size band could have the same a weight.

The calibration 'g' weight is based on auxiliary data and acts as a control on the 'a' weight to ensure that the final weight w is not skewed, based on over- or under-sampling businesses of a certain characteristic.

For example, suppose the 'a' strata are created based on 2 digit SIC groups. If one third of businesses respond, it follows that the responders have weights of 3 (ie: $N/n = 3/1 = 3$).

Furthermore, if the businesses that responded accounted for, say, 80% of the IDBR selected employment for that a group, weighting results by 3 is likely to produce employment and related variables of $3 \times 80\% = 240\%$ of the true value.

Therefore, the final weight (w) for each business in NIABI is:

$$w_i = a_i g_i$$

Also, the sum of (the control variable X weight) equals the sum of that calibration groups control variable. In the NIABI, the control variable is IDBR selected employment.

In the NIABI data files, the design weight ('a' weight) variable is called *aweight* (in the NIETS dataset this is called *dweight*); the 'g' weight variable is called *gweight* and the overall weight variable is called *calwght*.

- **Reporting Unit vs Local Unit**

An enterprise **reporting unit reports for all the local units within the enterprise** (for example, the reporting unit for a large supermarket chain will respond with aggregate figures incorporating all its Northern Ireland shops).

A local unit is an individual site (factory, shop, office, etc.) at which an enterprise conducts its business. For example, a supermarket chain may have shops in Newtownards, Bangor, Limavady, etc, these are local units

The **Annual Business Inquiry (including NIETS) dataset is at Reporting Unit level.**

A list of variables is included in [Annex A](#).

2.2 Northern Ireland Economic Trade Statistics (NIETS)

The Northern Ireland Economic Trade Statistics (NIETS), formerly known as the Broad Economy Sales and Exports Statistics (BESES), is a National Statistic dataset and provides an annual measure of local businesses' purchases and sales, to and from markets outside Northern Ireland (NI).

NIETS data is gathered through the Northern Ireland Annual Business Inquiry. In survey year 2011, questions relating to the value of trade were added to the NIABI questionnaire. This trade data (NIETS) is treated as a stand-alone dataset.

As the NIETS is a subset of the ABI, the salient points listed for the ABI also apply to the NIETS.

Like NIABI, the NIETS results are classified according to the Standard Industrial Classification of Economic Activities (SIC) system. Due to ongoing volatility of the estimates, both the NIABI and NIETS publications exclude estimates for Sector K (Insurance & Re-insurance industries. The other main areas that are excluded are: Public Administration and Defence (section O) while Agriculture, Forestry and Fishing (section A) excludes farming (groups 01.1, 01.2, 01.3, 01.4 and 01.5). Local authority and central Government bodies in Education (section P) and Human Health and Social Work Activities (section Q) have also been excluded from this publication, as has 86.2 (Medical and Dental Practice Activities) within section Q.

Other salient points include:

- The survey simply captures the value of a transaction and the destination of the sale.
- No information is collected on:
 - the business making the purchase;
 - the products sold;
 - whether the goods are final products or are intermediate goods to be used in a further production process;
 - the ultimate destination of any goods sold on; and
 - transport/shipping routes.

For further background information and details of the strengths and limitations of the NIETS data please see NIETS Quick Reference Guide, a link to which is available at the end of this document.

A list of variables is included in [Annex B](#).

2.3 Business Register and Employment Survey (BRES)

The Business Register and Employment Survey (BRES) is a survey of businesses in Northern Ireland. BRES is a compulsory government inquiry under Article 5 of the Statistics of Trade and Employment (Northern Ireland) Order 1988, which creates a legal obligation on businesses to make a return. The purpose of the BRES survey is two-fold:

1. Provide employee job figures for Northern Ireland; and
2. Update the NI element of the UK business register, known as the Inter-Departmental Business Register (IDBR).

The BRES collects employee job figures by gender and working pattern from public and private sector businesses in NI. The data allows for a disaggregation of the employee job counts by NI geographies as well as by industry. Information collected by the BRES (contact details including email address and business address, business operational status and business activity) is used to update the NI element of the IDBR which is used as a sampling frame for business surveys.

Salient points to note include:

- **Sampling Frame**

The Inter Departmental Business Register (IDBR) is the sampling frame used for all NISRA business surveys, including BRES. The register consists of companies, partnerships, sole proprietors, public authorities, central government departments, local authorities and non-profit making bodies in the UK.

BRES sample size and returns from 2014 to 2022 are as follows:

Year	2014	2015	2016	2017	2018	2019	2020	2021	2022
BRES									
Record numbers	12,000 /	55,000/ 49,970	12,000/ 11,106	30,000/ 28,058	12,000/ 10,449	33,500/ 25,924	16,000 /	33,500 /	34,800 /
(Approximate sample size/returns)	10,672						12,136	27,429	27,404

- **Stratification of the BRES sample**

The BRES sample consists of a fully enumerated census element and a stratified random sample of the remaining businesses. All public sector businesses are surveyed every year as are all businesses that have two or more local units and those with a reporting unit registered outside NI. The census element also includes all businesses with 6 or more employees for a larger sample and 20 or more employees for a smaller sample. The non-census element businesses are stratified by industry and geography levels and Neyman is applied to select the number of businesses within the strata required to provide a specified precision.

The sample design for BRES selects businesses based on employees within District Council area level and by Section and/or 2-digit SIC depending on the size of the sample required. BRES sampling is conducted at **Local Unit level**.

- **Sample size**

The BRES sample size changes year-on-year from 'big' to 'small' where a big BRES has a sample size in the order of 34,000 businesses in recent years and a small BRES has in the order of 12,000 businesses.

- **Imputation and Grossing**

BRES census element non-response is imputed. Imputation uses the most recent survey data available and a growth factor within strata at local unit level. BRES non-census element data are grossed using a statistical process called Generalised Estimation System (GES).

Please note that no imputation markers are available in the dataset.

- **Classifications**

The UK Standard Industrial Classification of Economic Activities 2007 (SIC 2007) is the classification framework used to define the industry in which a business operates. This is used by all business surveys within the UK.

A business may operate from one or more locations. The main business site or headquarters is referred to as a Reporting Unit. Individual sites (for example a factory or a shop) are called Local Units. BRES data is collected at local unit level.

- **Exclusions - BRES excludes the following:**

- All employees based outside NI;
- Agency workers paid directly from the agency payroll (n/a to recruitment agencies);
- Voluntary workers;
- Former employees only receiving a pension;
- Self-employed workers
- Working owners who are not paid via PAYE.

- Agriculture (but includes animal husbandry service activities and hunting, trapping and game propagation);
- HM armed forces;
- Private domestic servants; and
- Jobskills trainees without a contract of employment (non-employed status).

- **Blanks And Missing data**

The BRES weighting process is extremely complex and requires specialist software to undertake this. This software will not allow non-zero values or blank/missing variables to exist or else the algorithms will not work. Therefore, all cells in the database are numerated which means that it is not possible to leave a cell blank if a business did not provide data for a particular variable. Therefore, care is required when using the data for unit/firm level analysis as some zeros could actually be blanks.

- **Weights**

The use of weights within survey data is important to help ensure that the data is representative of the overall population. Users should note that the BRES dataset includes weight variables, which have been calculated specifically for the sample drawn for each survey cycle, and used in the production of the BRES published results. When replicating the published BRES figures it is important that the correct weight is used, i.e. the “calweight”.

- **Reporting Unit vs Local Unit**

An enterprise **reporting unit reports for all the local units within the enterprise** (for example, the reporting unit for a large supermarket chain will respond with figures incorporating all its Northern Ireland shops).

A local unit is an individual site (factory, shop, office, etc.) at which an enterprise conducts its business. For example, a supermarket chain may have shops in Newtownards, Bangor, Limavady, etc, these are local units

The **Business Register and Employment dataset is at Local Unit level**.

- **BRES Reporting Unit (RU) Variables**

NI BRES data is sampled, cleaned, imputed, grossed and published at Local Unit (LU) Level.

The weighted LU employee job data are aggregated to LU industry and geography levels and used to produce the NI BRES annual employee job national statistics, published by NISRA.

The RU industry and geography variables on the BRES datasets are only used by NISRA in the data collection process.

Researchers are advised to note the following:

- **RU District Council Area:** The RU District Council Area has been determined by matching the RU postcode with the NI Central Postcode Directory (CPD). The RU postcode is the postcode of the address provided by the business for data collection purposes. In many cases, this will be the head office but may also be a PO Box, home address, or the address of a solicitor or accountant. The RU District Council Area will be blank where a match with the CPD was not possible, such as for non-NI addresses.
- Published data at geographical breakdowns may not match perfectly due to variations in postcode allocations on the NI Central Postcode Directory (CPD) and continual validation work on the BRES annual survey data files.
- **RU SIC Codes:** The RU industry variables relate to the SIC code attributed to an RU when the business was selected for survey. An RU SIC is the dominant SIC when LU employee jobs are summed at 5-digit SIC level.

A list of variables is included in [Annex C](#).

2.4 Annual Survey of Hours and Earnings (ASHE)

The Annual Survey of Hours and Earnings (ASHE) for Northern Ireland is conducted on a statutory basis under the Statistics of Trade and Employment (Northern Ireland) Order 1988 (Article 5). It collects data on earnings by gender, work patterns, industry, occupation, and public vs. private sector pay comparisons.

The ASHE is a National Statistic³ dataset comprising approximately 1% of all employees in NI who were covered by Pay As You Earn (PAYE) schemes.

Salient points to note around the dataset include:

- **Sampling Frame**

ASHE sample size and returns from 2014 to 2022 are as follows:

Year	2014	2015	2016	2017	2018	2019	2020	2021	2022
ASHE									
Record numbers (sample size/returns)	3,091/ 2,311	3,147/ 2,293	3,114/ 2,375	3,118/ 2,378	3,196/ 2,266	3,249/ 2,397	3,191/ 2,195	3,155/ 2,248	3,310/ 2,378

- **Sample size**

³ National Statistics status means that our statistics meet the highest standards of trustworthiness, quality, and public value, and it is our responsibility to maintain compliance with these standards. National Statistics accreditation is granted by the UK statistics Authority following comprehensive and independent assessment of the statistics.

The ASHE sample includes approximately 8,000 jobs across 3,000 businesses per year.

- **Classifications**

The UK Standard Industrial Classification of Economic Activities 2007 (SIC 2007) is the classification framework used to define the industry in which a business operates. This is used by all business surveys within the UK.

- **Inclusions and exclusions**

The ASHE dataset contains information on approximately 5,700 jobs in all industries and occupations. It does not cover the self-employed. Given the survey reference date in April, the survey does not fully cover certain types of seasonal work, for example, employees taken on for only summer or winter work.

- **Weights**

Returned data are weighted to NI population totals from the Labour Force Survey (LFS) based on classes defined by occupation, region, age and sex. There are two processes involved in the weighting of responses for ASHE.

The first allocates individual cases a design weight to adjust for non-response. For this purpose, responses are treated as being in one of four strata, depending on whether they were part of the original questionnaire despatch, one of the later supplementary surveys or have a special arrangement in place with ONS to return their data electronically.

For the second part of the weighting, the final file of responses is post-stratified to population estimates taken from the LFS in 108 post-strata. These post-strata are defined as a cross-classification of:

- occupation (nine groups) – major groups from [Standard Occupational Classification 2010](#)
- age-band (three groups) – 16 to 21 years, 22 to 49 years and 50 years and over
- sex (two groups) – male and female
- region (two groups) – London and South East, and the rest of the UK

In order to produce estimates for the number of jobs falling below the National Minimum Wage, the dataset is re-weighted to exclude employees whose earnings were affected by absence during the reference pay period.

Users should note that weight variables are used in the production of the ASHE published tables. In order to replicate published ASHE estimates, the weighting variable 'calwght' should be used. The only exception to this is when estimating numbers of jobs with pay below the NLW/NMW or living wage, for which 'lpcalwghtf' should be used for 2021 and 'lpcalwght' for 2022.

It is also important to include the correct filters. Below is an example of the use of weights and filters to calculate summary figures for weekly pay-gross using STATA code.

```
summarize gpay [aweight = calwght] if (adr==1 & lop==2 & age>15), detail.
```

For more information on weights and filters please refer to the [ASHE dataset notes](#).

- **Job vs Individual vs Reporting Unit vs Local Unit**

The **ASHE dataset provides information at job and individual employee levels**. While the ASHE dataset includes the Anonymised_RURef variable, this variable is provided to allow the ASHE dataset to be linked to the other BDR datasets. The ASHE dataset does not include information at

Reporting Unit level but rather SIC and geographical data relate to the individual business (Local Unit) which employs the individual employee sampled within the data.

A list of variables is included in [Annex D](#).

2.5 Linkage Assessment

2.5.1 NIABI and BRES

An assessment of the linkage of NIABI and BRES was conducted to check the viability of linking the datasets over the years 2014 to 2020 and to highlight any potential issues that researchers should be aware of.

The work showed that it is indeed feasible to link NIABI and BRES datasets as each holds a unique reference number to identify the business. This number is the Reporting Unit Reference (RURef) and it allows an unambiguous match between businesses on the NIABI and BRES surveys across the years. That is to say, the businesses either match exactly by RURef or they don't. Therefore, the idea of false positives or false negatives for the NIABI/BRES matching does not apply and match accuracy will always be 100%.

Please note that whilst both the NIABI and BRES datasets are linkable by Reporting Unit reference, issues can arise given the caveats stated below. Any linkage between the BRES and NIABI datasets should be undertaken with caution as the datasets are designed at different levels for different purposes.

Essentially the NIABI/NIETS survey is designed at the RU level, so its survey forms are sent to the business headquarters and collect information at the business headquarters level.

The BRES survey forms are also sent to the business headquarters but collect information at the local unit level, i.e. information for the individual sites under the business headquarters.

This difference means the survey sample design is not the same for NIABI/NIETS compared to BRES. So NIABI/NIETS offers robust statistical results at the RU level and BRES offers robust statistical results at the LU level.

NIABI to BRES match rates (RU matches as a proportion of all NIABI RUs) were shown to be 45.2% in 2014, 96.9% in 2015, 54.6% in 2016, 80.2% in 2017, 57.8% in 2018, 82.8% in 2019 and 57.7% in 2020.

NIABI to BRES RU match rates are higher in years 2015, 2017 and 2019 because NIABI was matched against a much larger BRES sample in these years, therefore increasing the chance of a match. Otherwise, the average match rate was around 53.7%.

BRES to NIABI match rates (RU matches as a proportion of all BRES RUs) were shown to be 31.6% in 2014, 13.6% in 2015, 29.7% in 2016, 16.3% in 2017, 35.4% in 2018, 18.3% in 2019 and 31.7% in 2020.

BRES to NIABI RU match rates were lower in years 2015, 2017 and 2019 as these years had a much larger BRES sample, so the match rate denominator was much larger. Otherwise, the average match rate was around 31.6%.

The NIABI BRES linkages are subject to the following considerations for researchers:

- Linkage rates across the years, i.e. the proportion of matches achieved longitudinally for each of the surveys. These can be affected by a number of factors including:
 - the use of sample data within the BDR surveys such that not all businesses will appear in every year;
 - the loss of businesses over time due to, for example, the closure or splitting up of businesses;
 - the addition of new businesses over time and the changes to businesses due to mergers etc.; and
 - the level of consistency of data across the years – any known changes over time.

The match rate proportions were also broken down by SIC2, Industry Section, Legal Status and LGD and compared with the original NIABI and BRES proportions for each year. They were all found to be consistent.

Having conducted these detailed checks both RSU and ELMS are content that the NIABI and BRES data pass the quality assessment for linkage and they are fit to be linked in future projects.

2.5.2 ASHE and NIABI and BRES

Similar assessments of the linkage of ASHE and NIABI and the linkage of ASHE and BRES were conducted to check the viability of linking the datasets over the years 2014 to 2022.

Analysis has been carried out on what proportion of ASHE, ABI and BRES data, held in the BDR, is common for the years 2014 to 2022. Each survey begins with sampling from the Inter-Departmental Business Register (IDBR). This enables exact linkage on a common, and static, unique identifier. Each Reporting Unit (RU) on the IDBR has a RU reference (RURef) and the RU refers to the totality of the business' economic activity solely within Northern Ireland.

RU linkage: ASHE, ABI and BRES were matched by RURef for the years 2014 to 2022 and overall match rates were examined for each year. Given that the matching is exact i.e. there is no linkage error, when we discuss match rates we are really discussing proportions of samples which overlap on each survey. Each survey has a particular sample design which has been developed to maximise the accuracy of the survey specific outputs. A low match rate does not mean poor linkage but rather low levels of commonality of survey response data.

Year	ABI RU Count	ASHE RU Count	RU Match Count	ABI RU Match %	ASHE RU Match %
2014	7,445	2,311	1,134	15.2%	49.1%
2015	6,999	2,293	1,109	15.9%	48.4%
2016	6,044	2,375	1,066	17.6%	44.9%
2017	5,713	2,378	995	17.4%	41.8%
2018	6,409	2,266	1,175	18.3%	51.9%
2019	5,770	2,397	1,151	20.0%	48.0%
2020	6,664	2,195	1,098	16.5%	50.0%
2021	6,034	2,248	1,070	17.7%	47.6%
2022	6,614	2,378	1,164	17.6%	49.0%

Year	BRES RU Count	ASHE RU Count	RU Match Count	BRES RU Match %	ASHE RU Match %
2014	10,672	2,311	1,487	13.9%	64.3%
2015	49,970	2,293	2,215	4.4%	96.6%
2016	11,106	2,375	1,545	13.9%	65.1%
2017	28,058	2,378	2,046	7.3%	86.0%
2018	10,449	2,266	1,516	14.5%	66.9%
2019	25,924	2,397	2,006	7.7%	83.7%
2020	12,136	2,195	1,386	11.4%	63.1%
2021	27,429	2,248	1,871	6.8%	83.2%
2022	27,404	2,378	1,959	7.2%	82.4%

ASHE Longitudinal Jobs: Individuals selected for the ASHE survey are typically included for life, although there are a few exceptions. This means that there is a strong overlap between consecutive years of the ASHE samples, and this was shown in the longitudinal ASHE data. Over half (53.2%) of ASHE jobs sampled in 2022 were also present in the 2014 ASHE sample. This percentage increases steadily year by year with almost 76% of ASHE sampled in 2022 also being in the 2021 ASHE sample. There is a strong overlap between consecutive years of the ASHE which degrades over time.

Having conducted these detailed checks both RSU and ELMS are content that the ASHE, ABI and BRES data pass the quality assessment for linkage and they are fit to be linked in future projects.

Users should note that, as part of ongoing quality assurance of the IDBR, it has emerged that some RU references change over time. This is occurring for various reasons but in order to enable longitudinal analysis a table has been produced that links the RURef affected and the corrected RURef, per year.

The table is called [IDBR Linking RUREfs Anon] and will be made available to all approved research teams using the BDR. It contains 3 variables, which are Anon_RURef (Anonymised RURef), Year and Anon_Link_RURef (Anonymised Linking RURef). The table can be joined to the existing ASHE, NIABI, NIETS and BRES tables by joining Year and Anonymised_RURef to Anon_RURef. The Anon_Link_RURef should be used for any longitudinal analysis going forward to ensure consistency in the RUREfs over time.

3 Data access

The BDR data may be accessed by accredited researchers who have successfully submitted a project proposal to RSU. RSU will assist researchers from the beginning of their application proposal to the close of the project. Included in the proposal will be a variable list and researchers are required to justify the inclusion of each variable within the scope of their projects aims and objectives. Researchers are also required to detail any public engagement activities planned as part of their project. In addition, all projects must be accredited by the UK Statistics Authority Research Accreditation Panel (RAP) and have ethical approval. ELMS will give final approval to all projects.

Upon approval by RAP and ELMS, researchers will be provided with a suite of linkable tables supplied as flat files. The data will be made available to researchers in NISRA's secure environment or via the Office for National Statistics (ONS) Secure Research Service (SRS).

4 Acknowledgements

When publishing or presenting findings from the BDR the following acknowledgement should be used:

Administrative Data Research - Northern Ireland (ADR-NI) takes privacy protection very seriously. All information that directly identifies individuals/organisations will be removed from the datasets by trusted third parties before researchers access them.

All researchers using ADR-NI are trained and accredited to use sensitive data safely and ethically, they will only access the data via a secure environment and all their findings will be vetted to ensure they comply with strict confidentiality requirements.

The help provided by the staff of the Administrative Data Research Centre - Northern Ireland (ADRC-NI) and the ADR-NI support officers within NISRA Research Support Unit (RSU) is acknowledged. ADR-NI is funded by the Economic and Social Research Council (ESRC). The authors alone are responsible for the interpretation of the data and any views or opinions presented are solely those of the author and do not necessarily represent those of ADR-NI. The NISRA ELMS data has been supplied for the sole purpose of this project.

Annex A: NIABI Variable Catalogue

Note that financial values are in £'000 whilst employment figures are actuals.

Variable Name	Variable Description	Comments	Further Information
Anonymised_RURef	Anonymised Reporting Unit Ref	Anonymised business reference number for linking	
Year	Year		
Legalst	Legal Status		Origins in IDBR survey, for more information see Annex E for more information on the IDBR sources.
Ruemp	Reporting Unit Employees		
Ruempmt	Reporting Unit Employment		
Seltover	Selected Turnover		
Psic	Population SIC		For origins of SIC codes see Annex E.
live_LU	Num Live LU		Origins in IDBR survey, for more information see Annex E for more information on the IDBR sources.
WQ050	Total Employees		
WQ051	Total number of part-time employees		
WQ052	Total number of full-time employees		
WQ055	No. Working props, partners, exec directors		
WQ058	Unpaid workers, excluding voluntary workers		
WQ059	Total of 50, 55 and 58		
WQ317	Value of insurance claims received		
WQ318	Value of grants, donations, legacies, etc.		
WQ321	VAT included in total turnover		
WQ346	Total Turnover (including VAT)		
WQ360	Construction activity sub-contracted from another firm		
WQ362	Construction activity carried out outside Northern Ireland		
WQ399	Total turnover		
WQ400	Total taxes and levies paid		Note: WQ412+WQ413+ WQ15+WQ431 does not always sum to WQ400

Variable Name	Variable Description	Comments	Further Information
WQ412	Amount business rates (national non-domestic)		
WQ413	Other taxes, duties & levies (eg excise duties)		
WQ414	Total subsidies from government sources & EU		
WQ415	Amount paid for climate change levy		
WQ416	Total excise drawback & allowances from HMRC		
WQ431	Vehicle excise duty (road, car or vehicle tax)		
WQ450	Total employment costs		
WQ499	Total purchases of goods, material & services		Note: WQ710+WQ720+ WQ730 does not always sum to WQ499
WQ500	Total value of all stocks at beginning of period		
WQ501	Total value of work in progress - start of period		
WQ502	Total value of work in progress - end of period		
WQ505	Stocks at start of period-purchased for resale		
WQ506	Stocks at end of period-purchased for resale		
WQ550	Turnover excluding VAT		Note: WQ399+WQ346- (WQ321+WQ312+ WQ31]+WQ314) does not always sum to WQ550
WQ599	Total value of all stocks at end of the period		
WQ600	Cost of acquisitions		
WQ602	Capital nature work carried out by own staff		
WQ610	Acquisitions of land		
WQ611	Acquisitions of existing buildings		
WQ612	Acquisitions: Computer software developed by your own staff		
WQ613	Value of work carried out by own staff for business use		
WQ614	Acquisitions: Computer software purchased or developed externally		

Variable Name	Variable Description	Comments	Further Information
WQ615	Acquisitions: Machinery and equipment (previously plant)		
WQ616	Acquisitions of vehicles		
WQ617	Acquisitions of new buildings		
WQ618	Other construction		
WQ619	Computer hardware		
WQ620	Any other acquisitions		
WQ621	Assets acquired under finance leasing arrangements		
WQ622	Acquisitions: New construction work, new build		
WQ625	Acquisitions: Transfer costs and professional charges		
WQ626	Acquisitions: Natural resources		
WQ627	Acquisitions: Other non-produced assets		
WQ628	Acquisitions: Of which: Goodwill		
WQ629	Acquisitions: Intellectual property assets		
WQ630	Proceeds from the disposal of land		
WQ631	Proceeds from the disposal of existing buildings		
WQ632	Proceeds from any other disposals Include: machinery, equipment and vehicles		
WQ634	Proceeds from the disposal of plant, machinery and other capital equipment		
WQ635	Disposals: Transfer costs and professional charges		
WQ636	Disposals: Computer software developed by your own staff		
WQ637	Disposals: Computer software purchased or developed externally		
WQ638	Disposals: Machinery and equipment		

Variable Name	Variable Description	Comments	Further Information
WQ639	Disposals: Natural resources		
WQ640	Investments in concessions, patents, licences and trademarks		
WQ641	Disposals: Other non-produced assets		
WQ642	Disposals: Intellectual property assets		
WQ643	Mineral exploration and evaluation		
WQ644	Acquisitions: Contracts and fees for player transfers		
WQ645	Disposals: Contracts and fees for player transfers		
WQ690	Net capex (acquisitions - disposals)		Note: WQ600-WQ699 does not always sum to WQ690
WQ699	Proceeds from disposals		Note: WQ630+WQ631+ WQ636+ WQ637+WQ638+ WQ639+WQ641+ WQ642+WQ645 does not always sum to WQ699
WQ710	Total purchases of goods and materials		
WQ711	Of which for resale - NO further processing		
WQ712	Of which: were bought for further processing		
WQ713	Further Processing Purchases attributed to NI		
WQ714	Of which: semi-processed food/drink materials		
WQ716	Semi-Processed Purchases attributed to NI		
WQ717	Bought for Resale Purchases attributed to NI		
WQ720	Total purchases of energy and water		Note: WQ732+WQ729 does not always sum to WQ720
WQ721	Electricity		
WQ722	Natural Gas		
WQ723	LPG (liquid petroleum gas)		
WQ724	Heating Oil		
WQ725	Solid Fuels		
WQ726	Motor Fuels		

Variable Name	Variable Description	Comments	Further Information
WQ727	Other Fuel Energy Products		
WQ729	Energy Costs Total		Note: WQ721+WQ722+ WQ723+WQ724+ WQ725+WQ726+ WQ727 does not always sum to WQ729
WQ730	Total purchases of services		Note: WQ780+WQ770+ WQ771+WQ772+ WQ1830+WQ734+ WQ775 does not always sum to WQ730
WQ732	Of value at 7.2(b) how much was water charges		
WQ734	How much was sewerage & other disposal costs		
WQ770	Amounts payable for road transport services		
WQ771	Amounts payable for hiring, leasing or renting plant		
WQ772	Amounts payable for telecommunication services		
WQ775	All other amounts payable for services		
WQ780	Amounts payable to consultants or architects		
WQ810	Value of goods of own production		
WQ816	Construction & sales: own staff/sub-contractors		
WQ820	Work on customers materials/services rendered		
WQ830	Val gds purchas/resold NO more reprocessing		
WQ910	Turnover from sale of motor vehicles		
WQ911	T'ovr gds repurchas/resold NO more processing		
WQ920	Turnover from other motor trade activity		
WQ930	Turnover from non motor trade activity		
WQ1830	Agency Pay (Excluding VAT)		

Variable Name	Variable Description	Comments	Further Information
WQ2010	gvamp = GVA at market prices		Note: WQ550+WQ317+ WQ599+WQ502- WQ500- WQ501+WQ602- WQ499 does not always sum to WQ2010
WQ2020	gvafo = GVA at factor cost		Note: WQ2010+WQ414- WQ400+WQ416 does not always sum to WQ2020
WQ2030	gvabp = GVA at basic prices		Note: WQ2020+WQ412+ WQ431 does not always sum to WQ2030
rusection	Reporting Unit Section		
AWEIGHT	Survey Design Weight		
GWEIGHT	Survey G Weight		
CALWGHT	Survey Calibration Weight		
LGD_2014	Local Government District 2014		Based on BDR_GEOG CPD file joined by Year and Postcode.

Annex B: NIETS Variable Catalogue

Variable Name	Variable Description	Comments	Further Information
Anonymised_RURef	Anonymised Reporting Unit Ref	Anonymised business reference number for linking	
export	Total_Exports_Value		[MF5008]+[MF5009]+[MF5010]
WQ111	Of which Export of Goods to the Rest of EU		
WQ112	Export of Goods outside the UK		[WQ114]+[WQ111]+[WQ113]+[WQ116]+[WQ115]+[WQ117]
WQ113	Of which export of goods to ROW		
WQ114	Of which export of goods to ROI		
WQ115	Of which Export of Services to the Rest of EU		
WQ116	Of which export of services to ROI		
WQ117	Of which export of services to ROW		
WQ118	Freight & Insurance Costs assoc. with Exports		
WQ120	Total imports goods & services from outside UK		[MF5013]+[MF5014]+[MF5015] OR [WQ124]+[WQ126]+[WQ121]+[WQ125]+[WQ123]+[WQ127]
WQ121	Of which Import of Goods from the Rest of EU		
WQ122	Import of Goods from outside the UK		
WQ123	Of which Import of Goods from ROW		
WQ124	Of which Import of Goods from ROI		
WQ125	Of which Import of Services from Rest of EU		
WQ126	Of which Import of Services from ROI		
WQ127	Of which Import of Services from ROW		
WQ128	Freight & Insurance Costs assoc. with Imports		

Variable Name	Variable Description	Comments	Further Information
WQ163	Export of Services outside the UK		
WQ164	Import of Services from outside the UK		
WQ1600	Total goods and services purchased from Northern Ireland		[WQ1601]+[WQ1602]
WQ1601	Goods purchased from Northern Ireland		
WQ1602	Services purchased from Northern Ireland		
WQ1603	Total goods and services purchased from Great Britain		[WQ1604]+[WQ1605]
WQ1604	Goods purchased from Great Britain		
WQ1605	Services purchased from Great Britain		
MF5000	Goods and Services sold to NI		
MF5001	Goods and Services sold to NI, Of which were: Goods		
MF5002	Goods and Services sold to NI, Of which were: Services		
MF5003	Goods and Services sold to GB		
MF5004	Goods and Services sold to GB, Of which were: Goods		
MF5005	Goods and Services sold to GB, Of which were: Services		
MF5008	The value of exports of goods and services to Republic of Ireland		
MF5009	The value of exports of goods and services to Rest of European Union (EU)		
MF5010	The value of exports of goods and services to Rest of World		
MF5013	The value of imports of goods and services from Republic of Ireland		
MF5014	The value of imports of goods and services		

Variable Name	Variable Description	Comments	Further Information
	from Rest of European Union (EU)		
MF5015	The value of imports of goods and services from Rest of World		
MF3401	Germany		Researchers should be aware that analysing exports by country—particularly when combined with other variables such as SIC codes—will likely require aggregation to ensure the outputs are safe for release.
MF3402	France		
MF3403	Belgium		
MF3404	Luxembourg		
MF3405	Netherlands		
MF3406	Italy		
MF3407	Denmark		
MF3408	Portugal		
MF3409	Spain		
MF3410	Greece		
MF3411	Austria		
MF3412	Sweden		
MF3413	Finland		
MF3414	Cyprus		
MF3415	Czech Republic		
MF3416	Estonia		
MF3417	Hungary		
MF3418	Latvia		
MF3419	Lithuania		
MF3420	Malta		
MF3421	Poland		
MF3422	Slovakia		
MF3423	Slovenia		
MF3424	Bulgaria		
MF3425	Romania		
MF3426	Croatia		
MF3450	Rest of European Total		
MF3501	Switzerland		
MF3502	Turkey		
MF3503	Eastern Europe Incl Former USSR		
MF3504	Norway		
MF3505	Russia		
MF3598	EU Other Country 1 Sales		
MF3599	EU Other Country 2 Sales		
MF4101	United States Of America		
MF4102	Canada		
MF4103	Mexico		
MF4199	North America Other Country Sales		
MF4201	Brazil		

Variable Name	Variable Description	Comments	Further Information
MF4202	Argentina		
MF4203	Chile		
MF4298	Central & South America Other Country 1 Sales		
MF4299	Central & South America Other Country 2 Sales		
MF4301	Saudi Arabia		
MF4302	Israel		
MF4303	Jordan		
MF4304	Lebanon		
MF4306	Kuwait		
MF4307	Oman		
MF4308	Qatar		
MF4309	United Arab Emirates		
MF4310	Bahrain		
MF4398	Middle East Other Country 1 Sales		
MF4399/MF4305	Middle East Other Country 2 Sales (including Syria)		
MF4401	Japan		
MF4402	China		
MF4403	Hong Kong		
MF4404	India		
MF4405	Korea South		
MF4406	Singapore		
MF4407	Malaysia		
MF4408	Thailand		
MF4409	Indonesia		
MF4410	Philippines		
MF4411	Vietnam		
MF4412	Taiwan		
MF4498	Asia Other Country 1 Sales		
MF4499	Asia Other Country 2 Sales		
MF4501	South Africa		
MF4502	Egypt		
MF4598	Africa Other Country 1 Sales		
MF4599	Africa Other Country 2 Sales		
MF4601	Australia		
MF4701	New Zealand		
MF4850	Rest Of World Total		
MF4896	Rest of World Other Country 1 Sales		

Variable Name	Variable Description	Comments	Further Information
MF4897	Rest of World Other Country 2 Sales		
MF4898	Rest of World Other Country 3 Sales		
MF4899	Rest of World Other Country 4 Sales		
mic_s_m_lrg	Business Size (Employees)		

Annex C: BRES Variable Catalogue

Variable Name	Variable Description	Comments	Further Information
Year	Year of survey		
Anonymised_RURef	Anonymised Reporting Unit Ref	Anonymised business reference number for linking	Unique code identifies each business on the register. Each RURef can have one or more LURef. Rurefs are used by all business surveys in ELMS so would allow linkage.
Anonymised_LURef	Anonymised Local Unit Ref	Anonymised business reference number for linking over years	Unique code identifies each location within a business - so an RURef can have more than 1 LURef
LUEmpe	This is selected at employee count for the LU. LU_TOTAL_EMPS is the returned employee job count for that year		Local Unit employee jobs selected at value - i.e. at time selected for survey
LU_MALE_FT	Local Unit returned employee jobs Male FT		
LU_MALE_PT	Local Unit returned employee jobs Male PT		
LU_FEMALE_FT	Local Unit returned employee jobs Female FT		
LU_FEMALE_PT	Local Unit returned employee jobs Female PT		
LU_TOTAL_EMPS	Local Unit returned employee jobs Total		
LegalStatus	Legal Status	For more information see: Annex E	Note: Legal Status 4, 5 and 6 are public sector and the others private sector.
MALE_FT	RU MFT returned employee jobs	These are the returned employee jobs at RU level.	
MALE_PT	RU MPT returned employee jobs	These are the returned employee jobs at RU level.	
FEMALE_FT	RU FFT returned employee jobs	These are the returned employee jobs at RU level.	

Variable Name	Variable Description	Comments	Further Information
FEMALE_PT	RU FPT returned employee jobs	These are the returned employee jobs at RU level.	
TOTAL_EMPS	RU total returned employee jobs	These are the returned employee jobs at RU level.	
RUSic07	RU Level 5-digit SIC code	See Annex E UK SIC 2007 for info	RU 5-digit SIC code – as reported on the BRES survey form. Researchers are asked to note that LU SICs may differ to RU SIC.
RUSic07Desc	RU Level description of SIC code	See Annex E UK SIC 2007 for info	RU 5-digit SIC code description for the SIC_2007 variable (based on descriptions published by Companies House in Standard industrial classification of economic activities (SIC) - GOV.UK (www.gov.uk))
TIME_TAKEN	Time taken to complete the survey		Refers to survey burden.
LOCATION_COUNT	Count of local units per reporting unit		
w_mft	LU MFT employee jobs weighted		
w_mpt	LU MPT employee jobs weighted		
w_fft	LU FFT employee jobs weighted		
w_fpt	LU FPT employee jobs weighted		
w_total	LU TOTAL employee jobs weighted		
w_male	LU Male employee jobs weighted		
w_female	LU Female employee jobs weighted		
w_ft	LU Full-time employee jobs weighted		
Public_Private	Public Sector/Private Sector		
w_pt	LU Part-time employee jobs weighted		
Headline	Industry Headline		LU level
Section	Industry Section		LU level

Variable Name	Variable Description	Comments	Further Information
SIC2	2 digit code group	SIC2 - A sub-division of Industry_Section level. A 2-digit code (01 to 96).	LU level
SIC3	3 digit code group	SIC3 - A sub-division of SIC2. A 3-digit code (011 to 960)	LU level
SIC4	4 digit code group	SIC4 - A sub-division of SIC3. A 4-digit code (0111 to 9609)	LU level
SIC5	5 digit code group	SIC5 - A sub-division of SIC4. A 5-digit code (01110 to 96090)	LU level
LGD_2014_LU	Local Government District 2014 Local Unit	Based on CPD file available for each year of data.	LU Level
LGD_2014_RU	Local Government District 2014 Reporting Unit	Based on CPD file available for each year of data. Researchers are asked to note that LGD is based solely on the RU postcode on the form. Many of these will be non-NI, e.g. GB, ROI, overseas and will therefore have no RU LGD assigned to them and will be categorised as Unknown.	RU Level

Annex D: ASHE Variable Catalogue

Variable Name	Variable Description	Comments	Further Information
Year	Year for which the survey was conducted	The survey reference date is always in April	
Sex	Sex		
Djob	Double job marker	Indicates people with more than one job	
Mjob	Indicates the main job	Indicates the main job, but only for people with more than one job, 0 = Not main job, 1 = Main job	
Ft	Full Time / Part Time	Part time, full-time employees are defined as those who work more than 30 paid hours per week or those in teaching professions working 25 paid hours or more per week	
Empsta	Employee Start Date	Month and year in which employee started working for the organisation, MMYYYY	
occ10	Occupation Of Employee	Standard Occupational Classification 2010	Years 2014 - 2020
occ20	Occupation Of Employee	Standard Occupational Classification 2020	Years 2021 - 2022
Sjd	Same job	Has the employee worked in the same job for more than one year	
Pt	Permanent / temporary marker		
Payp	Pay period	Type of pay period that original returned data was based on. NB. all pay and paid hour variables in the dataset are already converted to weekly averages.	
Bpay	Basic Weekly Pay	Basic Weekly Earnings	
ovpay	Overtime Pay	Weekly overtime pay for the pay period	

Variable Name	Variable Description	Comments	Further Information
sppay	Shift Premium Pay	Additional premium payments during the pay period for shift work and night or weekend work not treated as overtime	
ipayall	Incentive Pay (2005)	Total incentive pay paid in this pay period, including any relating to another pay period	
ipayin	Incentive Pay In A Pay Period (2005)	Incentive pay that relates to this pay period	
othpay	Other Pay (2005)	Pay received in the pay period for other reasons	
tpay	Total Pay (2005)	Total pay as stated by respondent	
gpay	Weekly Earnings Including Overtime	bpay + ipayin + sppay + ovpay + othpay	
hexo	Hourly Earnings Excluding Overtime	gpox / bhr	<u>Note recorded in pence.</u>
he	Hourly Earnings Including Overtime	gpay / thrs	<u>Note recorded in pence.</u>
gpox	Weekly Earnings Excluding Overtime	Gross weekly earnings excluding overtime for the reference period	
hpay	Stated hourly rate of pay	Hourly rate of pay as stated by respondent	
lop	Earnings were affected by absence		
adr	Adult rate marker		
anipay	Annual Incentive Pay	Component of gross annual earnings that comes from incentive payments	
agp	Annual Earnings	Gross annual earnings paid for the tax year ending 5 April of the reference year	
bikfilter	Benefits in kind filter	Did the employee receive any benefits in kind in the tax year ending 5 April? For example, a company car or subsidised housing.	
Bik	Benefits In Kind (2005)	Annual value of benefits in kind for the tax year ending 5 April	

Variable Name	Variable Description	Comments	Further Information
Alday	Annual leave	Days	
Colag	Collective Agreement	Whether the employee's pay is set with reference to a collective agreement.	
Bhr	Normal Basic Weekly Hours	Basic weekly paid hours worked	
Ovhrs	Overtime Weekly Hours	Weekly paid overtime hours worked during the reference period	
Thrs	Total Weekly Hours	Total paid hours worked during the reference period bhr + ovhrs	
sic07	Standard Industrial Classification (2007)	Standard Industrial Classification 2007	
Age	Age	The dataset only contains people aged 16 and over at the survey reference date	<u>Bottom-coded at age 17 and top-coded at age 67.</u>
Harea	Home Local Government District 2014	Based on individual level home postcode	<u>Note some individuals with more than one job in a single year have different Harea recorded for different jobs. This may be due to a change of address not updated with all employers.</u>
Warea	Work Local Government District 2014	Based on work postcode	
Pubpriv	Public Private		
Bpayinc	Basic Pay including Other Pay	bpay + othpay	
Dweight	Survey Design Weight	Based on how the data was collected	
Gweight	Survey G Weight	Based on LFS population totals	
Calwght	Survey Calibration Weight	Weights used to calibrate ASHE returns to job totals from the Labour Force Survey based on a combination of dweight and gweight	
Lpdweight	Low pay design weight	Based on how the data was collected	
lpgweight	Low pay population weight	Based on LFS population totals	

Variable Name	Variable Description	Comments	Further Information
lpcalwght	Low pay calibration weight	Cases with loss of pay due to absence (or in industries T or U or unclassified) are firstly removed before calculating weights that sum to the number of jobs in the labour market	
ppfilter	Pension provision filter	Was the employee a member of any pension scheme run or facilitated by their organisation	
tpen	Type of pension scheme		
penpay	Pensionable Amount (2005)	The amount of employee's weekly pay that was pensionable	
ownpay	Employee Contribution To Pension (2005)	Weekly employee's pension contributions	
compay	Employer Contribution To Pension (2005)	Weekly employer's pension contributions	
ownperc	Employee's percentage pension contribution	The employee's contributions made in the pay period as a percentage of pensionable pay	
comperc	Employer's percentage pension contribution	The employer's contributions made in the pay period as a percentage of pensionable pay	
app	Apprentice marker	Apprentice marker	
appdate	Apprenticeship start date	Format 'MMYYYY'	
hrpayx	The derived hourly rate used in the low pay calculation	The derived hourly rate is hourly pay excluding overtime and shift premium pay	
Anonymised_piden	Anonymised person identifier	Anonymised version of a random identifier created to identify multiple occurrences of the same person within the dataset,	Unique code identifies each employee within a business - so an RURef can have more than 1 Person_Identifier

Variable Name	Variable Description	Comments	Further Information
		this allows comparison over time	
LOPF	Loss of pay but excluding furlough marker	Loss of pay but excluding furlough marker– only available for 2021	Available 2020 and 2021
Furlough	Furlough marker	Whether the employee has been furloughed on the COVID-19 Job Retention Scheme (CJRS) – only available for 2021 identified via Q2i	Available 2020 and 2021
Lpdweightf	Low pay calibration weight	Same as described for lpcalwghtf but based on stratum	Only available for 2021
Lpgweightf	Low pay calibration weight	Same as described for lpcalwghtf but based on LFS population totals stratum	Only available for 2021
Lpcalwgtf	Low pay calibration weight	The same as lpcalwght except that any cases with loss of pay due to absence who were furloughed are included (using Furlough). The aim is to produce weights similar to normal years' lpcalwght, i.e. excluding employees absent due to situations such as maternity/paternity leave, sickness etc.	Only available for 2021
Anonymised_RURef	Anonymised Reporting Unit Reference	Anonymised version for linking	
Idbrnemp	IDBR employment	Number of employees in the enterprise on the IDBR	<u>Please note that this information relates to the number of employees for a business at a UK rather than a NI level. NI employee information can be sourced from the other BDR datasets.</u>

Variable Name	Variable Description	Comments	Further Information
Stratum	Relates to the way in which the data was collected from 2004	Relates to the way in which the data was collected from 2004	
lowhc	Non minimum hours contract	Was the employee on a contract that did not guarantee a minimum number of hours work?	

Annex E: Useful online links

Please note that this User Guide and the BDR 2022 Metadata and Data Dictionary will be made available within the secure setting for researchers accessing the BDR data. Please contact rsu@nisra.gov.uk for further details.

[Code of Practice for Statistics](#)

Code of Practice for Statistics



Code-of-Practice-for-
Statistics-REVISED.pdf

[The Statistics of Trade and Employment \(NI\) Order 1988](#)

The Statistics of Trade and Employment (NI) Order 1988



The Statistics of
Trade and Employment

[ELMS guide to survey](#)

ELMS guide to survey



ELMS-guide-to-survey-2020 (1).XLSX

[ABI background information](#)

ABI Background Information



ABI background
information.docx

[Inter Departmental Business Register \(IDBR\)](#)

Inter Departmental Business Register (IDBR)



Inter Departmental
Business Register.doc

[ABI sample coverage](#)

ABI Sample Coverage



ABI sample
coverage.docx

[Usage, methods and quality and revisions policy - ABI](#)

Usage, Methods and Quality Revisions Policy – ABI



Usage Methods and
Quality Revisions Polic

[ABI survey form](#)

ABI survey form



ABI-long-form.pdf

[ABI questionnaire guidance notes](#)

ABI questionnaire guidance notes



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[NIETS background information](#)

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[NIETS Quick Reference Guide](#)

NIETS Quick Reference Guide



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[Further information about IDBR sources, structure and updating for publication](#)

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Further Information
About IDBR Sources.p

[UK SIC 2007](#)

UK SIC 2007



UK SIC 2007.pdf

[Info on Use of Weights in ABS Technical Report](#)

Information on Use of Weights in ABS Technical Report



ABS Technical Report
2012.pdf

[A Guide to Business Legal Status](#)

A Guide to Business Legal Status



A Guide to Business
Legal Status.pdf

[BRES guidance](#)

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[ASHE background quality report](#)



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[ASHE questionnaire guidance](#)



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aire-guidance-2023.

[ASHE sample form](#)



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Form.pdf

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[ASHE summary process diagram](#)



ASHE process
diagram.pdf

[ASHE methodology](#)



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[ASHE dataset notes](#)



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